

LD9 Glossary

Final Documentation Package for the Bank Marketing Subscription Prediction Project

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This glossary contains the main terms used across the Day 9 ensemble learning project.

Accuracy: The share of all predictions that were correct.

Bagging: An ensemble method that trains many models on bootstrap samples and combines their predictions.

Base learner: A first-level model used inside an ensemble system.

Blending: A simplified stacking approach that uses one train/blend split instead of full cross-validation.

Bootstrap sample: A sample made by drawing training rows with replacement.

Cross-validation: A method that evaluates a model across several folds for a more stable performance estimate.

Decision Tree: A tree-based model that makes predictions through a sequence of split rules.

Ensemble learning: Using multiple models together to produce a stronger final result.

F1-score: A metric that balances precision and recall.

Feature importance: A measure of how much a feature influenced a tree-based model.

Gradient Boosting: A sequential boosting method where each new tree corrects earlier errors.

KNN: K-Nearest Neighbors, a model that predicts based on nearby training cases.

Meta-model: A model that learns from the predictions of base models.

Model diversity: The idea that base learners should be different enough to contribute unique value.

OOF predictions: Out-of-Fold predictions created on rows that were not used to train the predicting model.

OOB score: Out-of-Bag score, an internal validation score often used in Random Forest.

Precision: Among predicted positive cases, how many were truly positive.

Random Forest: An ensemble of decision trees built with bagging and random feature selection.

Recall: Among the truly positive cases, how many the model correctly identified.

RSM: Random Subspace Method; using random subsets of features so trees become less similar.

ROC-AUC: A metric showing how well a model ranks positives above negatives.

Stacking: An ensemble method where a meta-model learns from the predictions of several base learners.

Three-layer stacking: A deeper stacking structure with base learners, intermediate meta-models, and a final meta-model.